

We want to study the relations between the dual group $\widehat{A}$ of A and the dual groups $\widehat{H}$ and $\widehat{A/H}$ of the subgroup H and the quotient group A/H . In fact, one can identify $\widehat{A/H}$ with the subgroup $H^\perp$, called the *annihilator* of H , defined as

$$H^\perp := \left\{ \chi \in \widehat{A} : \chi(x) = 1 \quad \forall x \in H \right\},$$

so that if we take $\phi \in L^1(A)$ and define $\phi^H \in L^1(A/H)$ as $\phi^H(xH) = \int_H \phi(xh)dh$, then by the above identification, we get $\mathcal{F}_{A/H}(\phi^H) = \mathcal{F}_A(\phi)|_{H^\perp}$, see the proof of the theorem below, which explains this technique.

Theorem 2.1 (General Poisson summation formula). *Let H be a closed subgroup of the locally compact Abelian group A . For $\phi \in L^1(A)$, if $\mathcal{F}_A(\phi)|_{H^\perp} \in L^1(H^\perp)$, then*

$$\int_H \phi(xh)dh = \int_{H^\perp} \mathcal{F}_A(\phi)(\chi)\chi(x)d\chi, \quad (2.3)$$

for all $x \in A$, where Haar measure on $H^\perp \cong \widehat{A/H}$ is the Plancherel measure with respect to the chosen Haar measure on A/H .

Proof. For $\chi \in H^\perp$ we have $\chi(xh) = \chi(x)$ for every $x \in A$ and $h \in H$. We therefore get from the quotient integral formula (2.2) that

$$\begin{aligned} \mathcal{F}_{A/H}(\phi^H)(\chi) &= \int_{A/H} \phi^H(xH) \overline{\chi(x)} d(xH) \\ &= \int_{A/H} \int_H \phi(xh) \overline{\chi(xh)} dh d(xH) \\ &= \int_A \phi(x) \overline{\chi(x)} dx = \mathcal{F}_A(\phi)(\chi) \end{aligned}$$

for every $\chi \in H^\perp$. Moreover, if $\mathcal{F}_A(\phi)|_{H^\perp} \in L^1(H^\perp) = L^1(\widehat{A/H})$, then the Fourier inversion formula implies that for all $x \in A$

$$\begin{aligned} \int_H \phi(xh)dh &= \phi^H(xH) = \mathcal{F}_{H^\perp} \mathcal{F}_{A/H}(\phi^H)(-xH) \\ &= \mathcal{F}_{H^\perp} \mathcal{F}_A(\phi)(-xH) = \int_{H^\perp} \mathcal{F}_A(\phi)(\chi) \overline{-xH(\chi)} d\chi \\ &= \int_{H^\perp} \mathcal{F}_A(\phi)(\chi) \overline{\chi(-x)} d\chi = \int_{H^\perp} \mathcal{F}_A(\phi)(\chi) \chi(x) d\chi. \end{aligned}$$

□

In all our applications, the group A will be of the form $A = (\mathbb{Z}/N\mathbb{Z})^{d_1} \times \mathbb{R}^{d_2}$ for $N \in \mathbb{N}_+$, $d_1, d_2 \in \mathbb{N}$. Recall we have defined the standard symmetric bilinear form on $\mathbb{R}^{d_2}$ as in the beginning of §1, and for any $N \in \mathbb{N}_+$, we still denote the symmetric bilinear form on $(\mathbb{Z}/N\mathbb{Z})^{d_1}$ by Q as defined in the standard way. We then identify A with its dual group $\widehat{A}$ through the pairing

$$\begin{aligned} A &\cong \widehat{A} \\ (s, t) &\mapsto \chi_{s,t}(l, \xi) = e^{2\pi i \left(\frac{Q(s,l)}{N} + Q(t,\xi) \right)}, \end{aligned}$$